Perturb-Seq Reanalysis Pipeline

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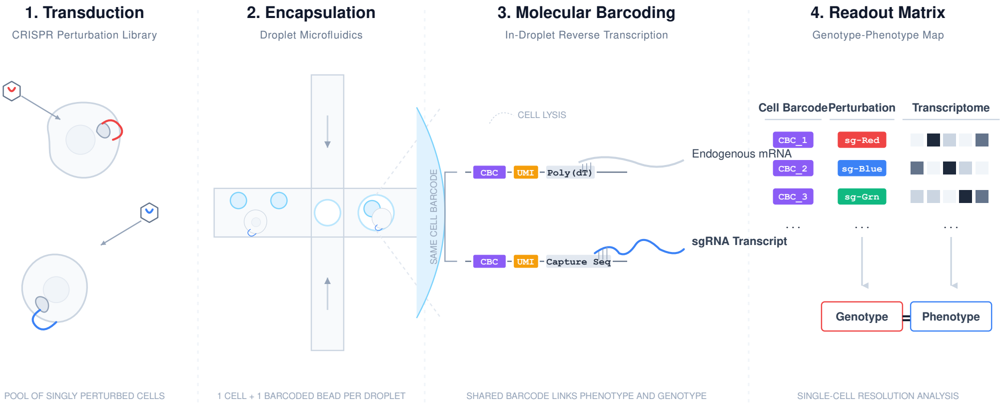
What is Perturb-Seq?



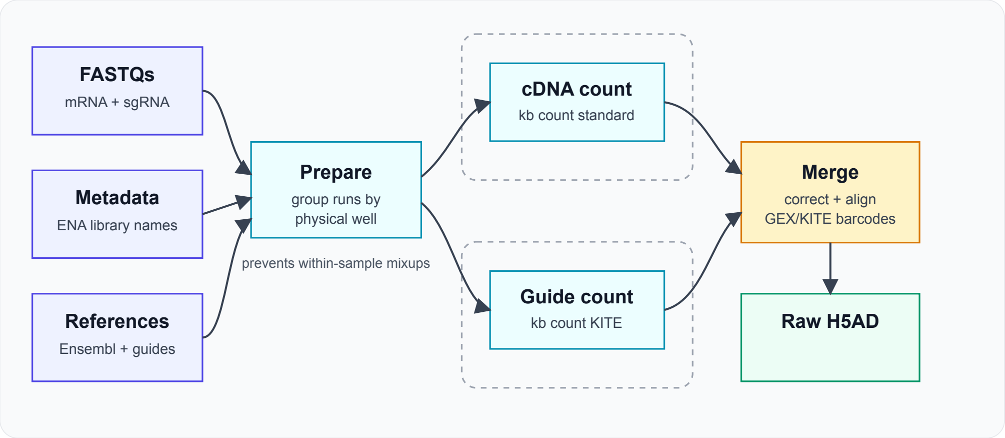
Why re-process from FASTQ?

- **Main problem:** scPerturb datasets are usually made available as processed H5AD files, but they come from different pipelines, references, and filtering choices.
- **Stable schema:** Re-processing gives us a common count matrix layout, common metadata expectations, and a consistent way to attach guide counts. This significantly simplifies downstream analysis.
- **Quality control:** Raw reads let us apply the same checks across studies instead of inheriting every author-specific decision.
- **Batch effects:** Reprocessing from scratch using a unified pipeline helps in reducing systematic batch effects between datasets.
- **Catalogue expectation:** Users expect harmonised data in the Perturbation Catalogue, not only links to heterogeneous processed files.

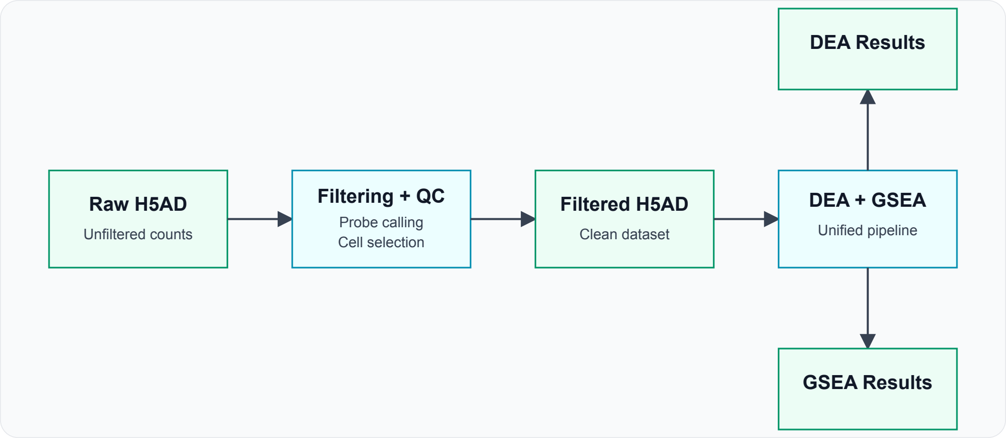
Why Kallisto + Bustools?

- **Speed:** Kallisto + Bustools is much faster than Cell Ranger while giving very comparable expression quantification.
- **Resource fit:** We have adequate, but not unlimited, compute and storage. The pipeline must work within those limits.
- **Full catalogue scale:** We want to process all datasets we have, and be able to reprocess them if the pipeline improves.
- **Future-proofing:** New Perturb-Seq datasets are expected to be very large, so the pipeline needs to scale well rather than depend on throwing more hardware at each run.

Pipeline Architecture



Downstream analysis



Challenges

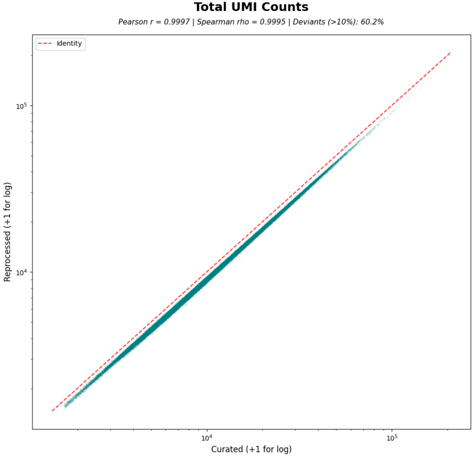
- **ENA FASTQs missing technical reads:** Historical ENA FASTQs for this dataset do not include the barcode/UMI reads. Current workaround: SRA download with technical reads included. We notified ENA; they are aware and plan to fix it, so future runs may use ENA directly.
- **Sample grouping:** SRRs and lanes must be grouped by physical well before counting. This is currently parsed from ENA library names; later versions may autodetect this.
- **Barcode collisions:** Final concatenation adds sample suffixes so identical 10x barcodes from different wells remain distinct.
- **GEX/KITE matching:** Guide barcodes are not exactly the same as expression barcodes in this dataset. The pipeline applies the required KITE barcode correction before alignment.

List of datasets

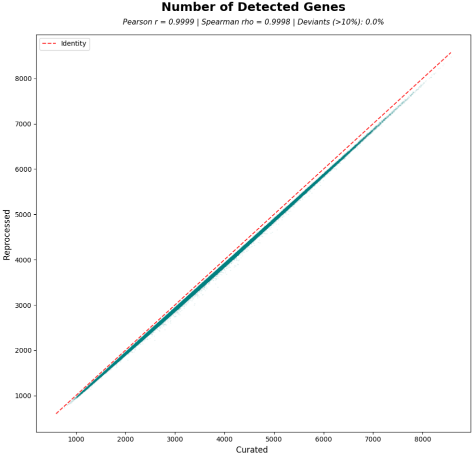
Dataset	Size (TB)	Total cells	Single-gene	Control	Multi-gene	No-guide
Nadig 2025 Jurkat	2.0	588,498	214,092	10,629	289,965	58,565
Nadig 2025 HepG2	2.6	501,121	129,909	4,835	333,994	21,803
Replogle 2022 K562 Essential	1.7	544,799	274,155	11,048	191,241	61,211
Replogle 2022 K562 GW	11.0	1,314,345	613,755	29,463	545,514	101,360
Replogle 2022 RPE1 Essential	1.9	509,962	206,740	11,855	197,064	79,695

- Currently, 5 datasets are processed.
- Gaussian-Poisson threshold used for guide calling (8-9 UMI).

Cell-Level Expression Agreement

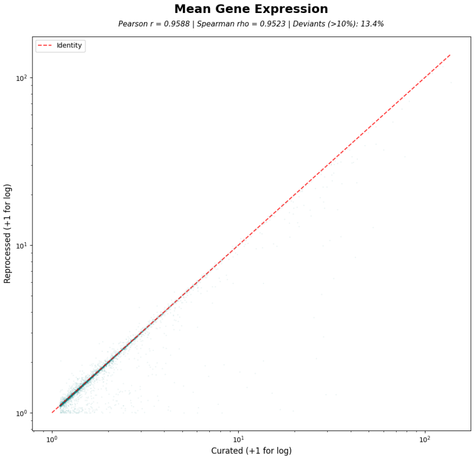


Correlation of total UMI counts per cell. Kallisto+Bustools (Reprocessed) generally identifies more UMIs per cell, especially in the high-sensitivity regime, likely due to transcript-level mapping vs gene-level quantification.

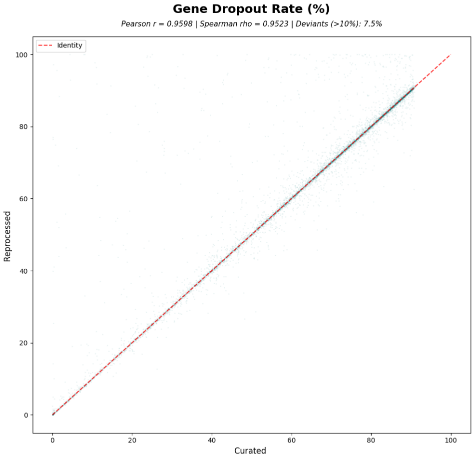


Correlation of unique genes detected per cell. The reprocessed pipeline recovers significantly more unique transcripts in a subset of cells, indicating higher detection sensitivity.

Gene-Level Expression Agreement



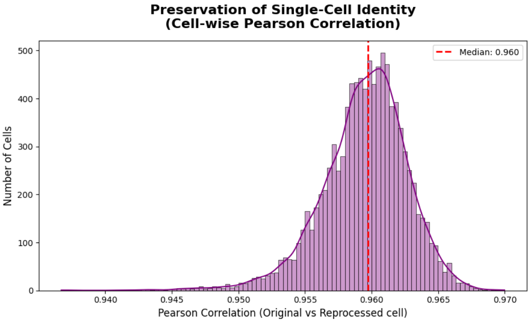
Average expression level for each shared gene. High correlation confirms that the biological signal is preserved across quantification methods.



Percentage of cells where a gene has zero counts. Lower values in Reprocessed indicate higher sensitivity for those genes.

Expression Vectors Are Preserved

- For each shared cell, we compare the original and reprocessed expression vectors over the same genes.
- The histogram shows the distribution of Pearson correlation values for a sample of 10,000 cells.



This distribution shows the correlation of the full expression profile for each individual cell against its exact counterpart in the reprocessed dataset. A strong peak near 1.0 confirms that single-cell identities are highly preserved.

Guide Recovery: Mixture Model Approach

- We use a **Gaussian-Poisson mixture model** to separate guide signal from background.
- The background is modeled as a Poisson distribution, while the true signal follows a Gaussian distribution.
- The decision threshold is automatically set at the crossover point where the probability of signal becomes higher than background.
- Cells are assigned a guide if at least one probe targets the gene and passes this threshold.
- $>$95% overall agreement on call outcomes between original and reprocessed data; for single gene cells, $>$99.8% agreement.

What We Have Now

- A working, reproducible FASTQ-to-H5AD pipeline for Perturb-Seq datasets.
- Both expression and guide calling are solved with very good alignment to original data.
- A comparison workflow that checks cell-level expression, gene-level expression, dropout, and perturbation labels.
- DEA and GSEA downstream analysis.
- Full pipeline was successfully ran for 5 datasets in the Catalogue.

Next Steps

- Further exploration and validation of results.
- Updating data in Catalogue and publishing a new release.
- Expand pipeline to deal with more datasets, 10x v2 chemistry, CSP data.
- Eventually: release pipeline as a separate package.